10/5/2026

Git-HUB

Artificial Intelligence Lab 05

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Class: BSCS F24 B



Lab No 05

Lab Task 1: Weather Logger

1. *(Easy)* **Implement** a Python script `weather_logger.py` that uses a `while` loop to repeatedly ask the user to enter a daily temperature (a number), stopping when the user types `"done"`; store the entered temperatures in a list. Then write a function `summarize(temps)` that **returns a dictionary** containing the `"minimum"`, `"maximum"`, and `"average"` temperature, and print the returned dictionary.

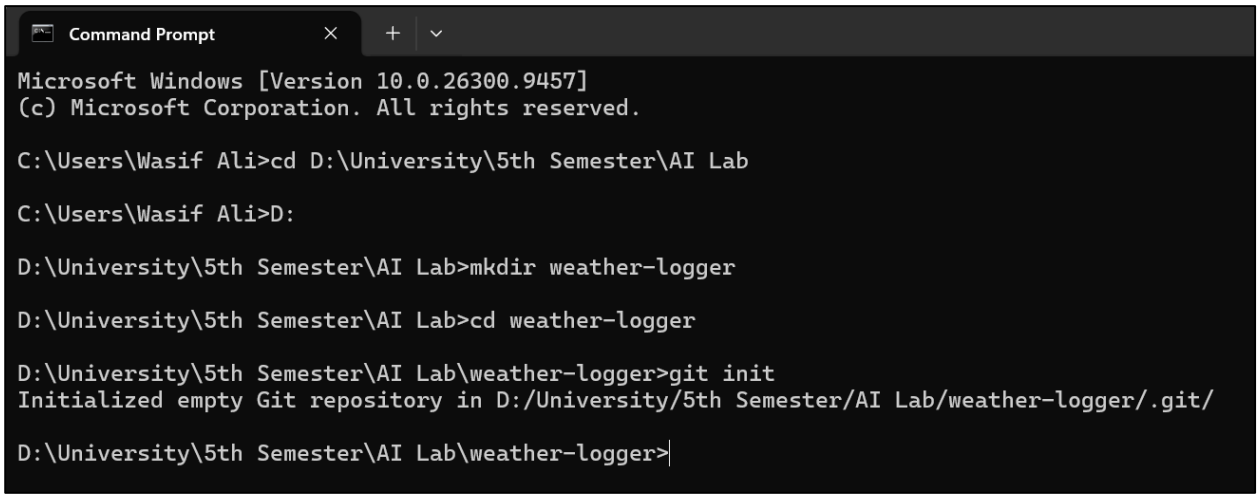
   Python – Sample Interactive Run:

   ```
   Enter temperature ('done' to stop): 28
   Enter temperature ('done' to stop): 31
   Enter temperature ('done' to stop): 25
   Enter temperature ('done' to stop): done
   {'minimum': 25, 'maximum': 31, 'average': 28.0}
   ```

   Mandatory Git/GitHub steps:

   (a) `git init` a repository named `weather-logger`.
   (b) Commit the first working version with `git add` and `git commit`.
   (c) Create and switch to a branch named `add-summary`, where you implement the `summarize()` function; commit this change on that branch.
   (d) Switch back to `main` and use `git merge` to bring in the `add-summary` branch.
   (e) Create a GitHub repository named `weather-logger` and push your local repository to it with `git remote add origin` and `git push`.

Create Directory and Initialize Git



Create Initial Python Script & Commit

weather_logger.py M

D: > University > 5th Semester > AI Lab > weather-logger > weather_logger.py > ...

```python
# weather_logger.py
temps = []
while True:
    val = input("Enter temperature ('done' to stop): ")
    if val.strip().lower() == "done":
        break
    temps.append(float(val))

print("Entered temperatures:", temps)
```

Stage and commit

```
D:\University\5th Semester\AI Lab\weather-logger>git add weather_logger.py

D:\University\5th Semester\AI Lab\weather-logger>git commit -m "Initial commit: basic temperature logging loop"
[master (root-commit) 2d85f48] Initial commit: basic temperature logging loop
 1 file changed, 9 insertions(+)
 create mode 100644 weather_logger.py

D:\University\5th Semester\AI Lab\weather-logger>
```

Branching & Feature Implementation

```
D:\University\5th Semester\AI Lab\weather-logger>git branch add-summary

D:\University\5th Semester\AI Lab\weather-logger>git switch add-summary
Switched to branch 'add-summary'

D:\University\5th Semester\AI Lab\weather-logger>
```

Update weather_logger.py to add summarize(temps)

weather_logger.py M

D: > University > 5th Semester > AI Lab > weather-logger > weather_logger.py > ...

```python
# weather_logger.py
def summarize(temps):
    return {
        "minimum": min(temps) if temps else 0,
        "maximum": max(temps) if temps else 0,
        "average": sum(temps) / len(temps) if temps else 0.0
    }

temps = []
while True:
    val = input("Enter temperature ('done' to stop): ")
    if val.strip().lower() == "done":
        break
    temps.append(float(val))

print(summarize(temps))
```

Commit on the branch

```
D:\University\5th Semester\AI Lab\weather-logger>git add weather_logger.py

D:\University\5th Semester\AI Lab\weather-logger>git commit -m "Add summarize function for min, max, and avg"
[add-summary d1aa354] Add summarize function for min, max, and avg
 1 file changed, 8 insertions(+), 1 deletion(-)

D:\University\5th Semester\AI Lab\weather-logger>
```

Merge Branch

```
D:\University\5th Semester\AI Lab\weather-logger>git switch master
Switched to branch 'master'

D:\University\5th Semester\AI Lab\weather-logger>git merge add-summary -m "Merge add-summary into master"
Updating 2d85f48..d1aa354
Fast-forward (no commit created; -m option ignored)
 weather_logger.py | 9 ++++++++-
 1 file changed, 8 insertions(+), 1 deletion(-)

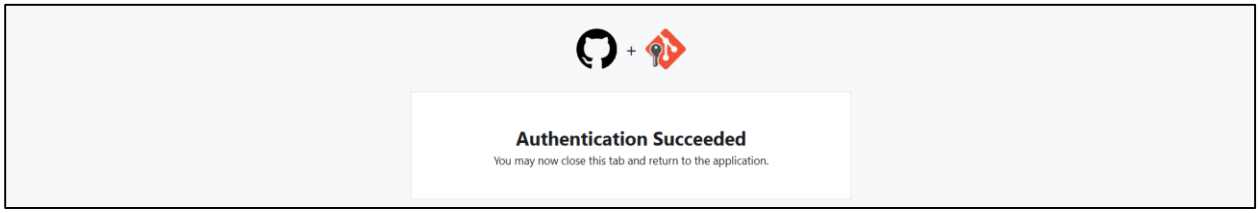
D:\University\5th Semester\AI Lab\weather-logger>
```

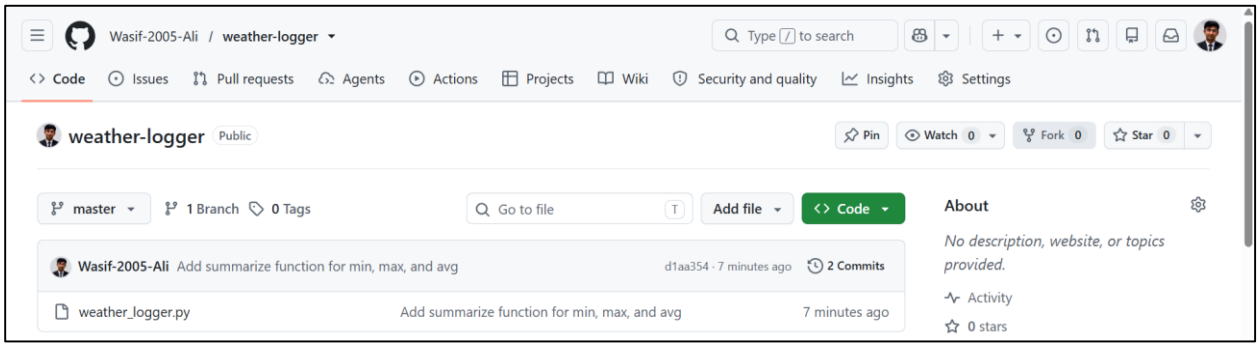
Push to GitHub

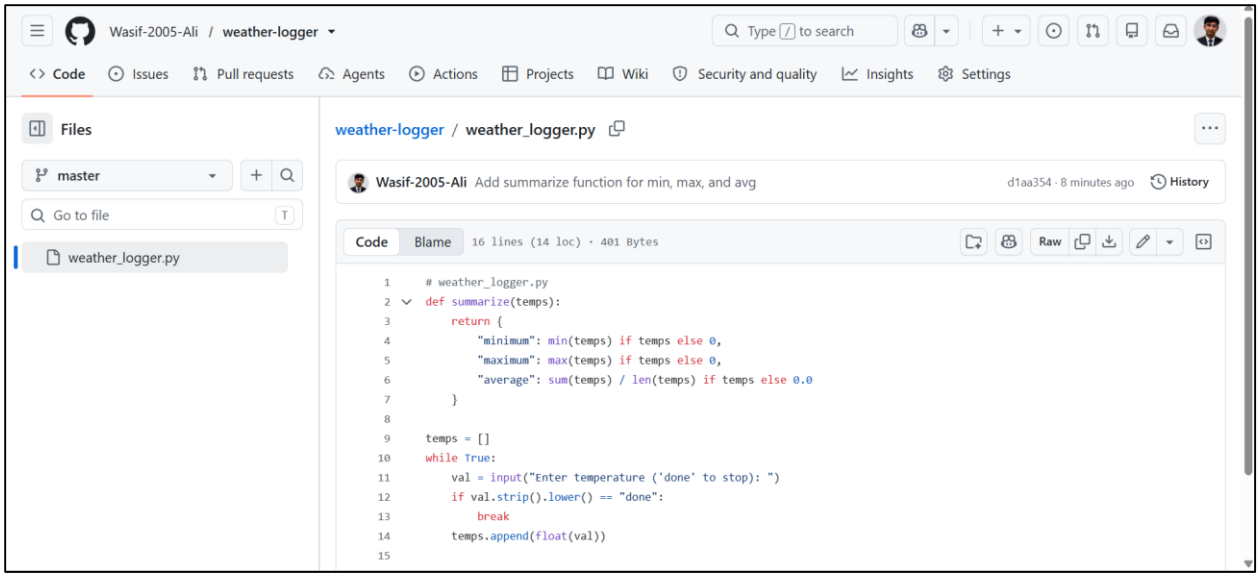
```
D:\University\5th Semester\AI Lab\weather-logger>git remote add origin https://github.com/Wasif-2005-Ali/weather-logger.
git

D:\University\5th Semester\AI Lab\weather-logger>git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 752 bytes | 752.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Wasif-2005-Ali/weather-logger.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

D:\University\5th Semester\AI Lab\weather-logger>
```







Lab Task 2: Shopping List

Create Directory and Initialize Git

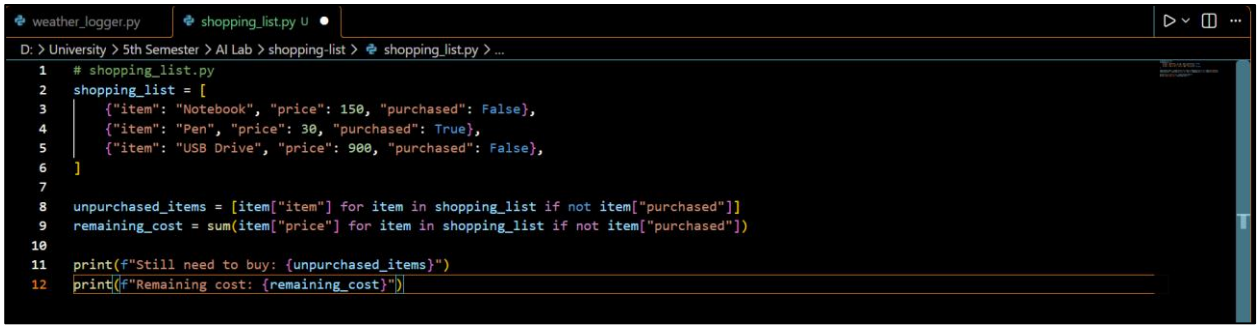
```
D:\University\5th Semester\AI Lab>mkdir shopping-list

D:\University\5th Semester\AI Lab>cd shopping-list

D:\University\5th Semester\AI Lab\shopping-list>git init
Initialized empty Git repository in D:/University/5th Semester/AI Lab/shopping-list/.git/

D:\University\5th Semester\AI Lab\shopping-list>
```

Create Initial Script & Commit



Stage and commit

```
D:\University\5th Semester\AI Lab\shopping-list>git add shopping_list.py

D:\University\5th Semester\AI Lab\shopping-list>git commit -m "Initial commit: shopping list comprehensions"
[master (root-commit) 07c9471] Initial commit: shopping list comprehensions
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 shopping_list.py

D:\University\5th Semester\AI Lab\shopping-list>
```

Modify File, Inspect Diff, and Commit

```
D:\University\5th Semester\AI Lab\shopping-list>git diff
diff --git a/shopping_list.py b/shopping_list.py
index e69de29..bfeedb8 100644
--- a/shopping_list.py
+++ b/shopping_list.py
@@ -0,0 +1,12 @@
+# shopping_list.py
+shopping_list = [
+    {"item": "Notebook", "price": 150, "purchased": True},
+    {"item": "Pen", "price": 30, "purchased": True},
+    {"item": "USB Drive", "price": 900, "purchased": False},
+]
+
+unpurchased_items = [item["item"] for item in shopping_list if not item["purchased"]]
+remaining_cost = sum(item["price"] for item in shopping_list if not item["purchased"])
+
+print(f"Still need to buy: {unpurchased_items}")
+print(f"Remaining cost: {remaining_cost}")
\ No newline at end of file

D:\University\5th Semester\AI Lab\shopping-list>
```

Stage and commit

```
D:\University\5th Semester\AI Lab\shopping-list>git add shopping_list.py

D:\University\5th Semester\AI Lab\shopping-list>git commit -m "Mark Notebook as purchased"
[master 876a665] Mark Notebook as purchased
 1 file changed, 12 insertions(+)
```

Record Commit History

```
D:\University\5th Semester\AI Lab\shopping-list>git log --oneline
876a665 (HEAD -> master) Mark Notebook as purchased
07c9471 Initial commit: shopping list comprehensions

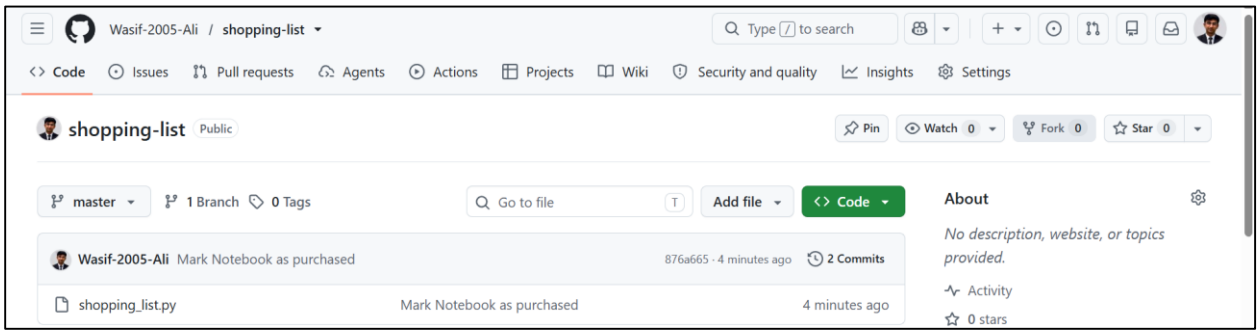
D:\University\5th Semester\AI Lab\shopping-list>
```

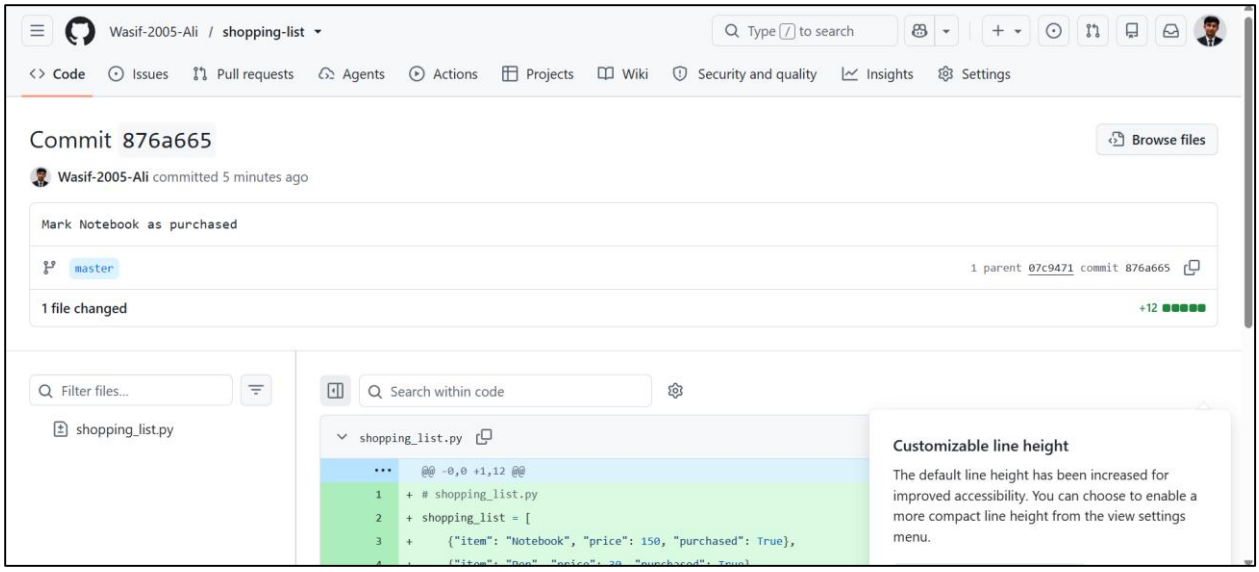
Push to GitHub

```
D:\University\5th Semester\AI Lab\shopping-list>git remote add origin https://github.com/Wasif-2005-Ali/shopping-list.gi
t

D:\University\5th Semester\AI Lab\shopping-list>git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 689 bytes | 689.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Wasif-2005-Ali/shopping-list.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

D:\University\5th Semester\AI Lab\shopping-list>
```





Lab Task 3: Habit Tracker

Create Directory and Initialize Git

```
D:\University\5th Semester\AI Lab\shopping-list>cd D:\University\5th Semester\AI Lab

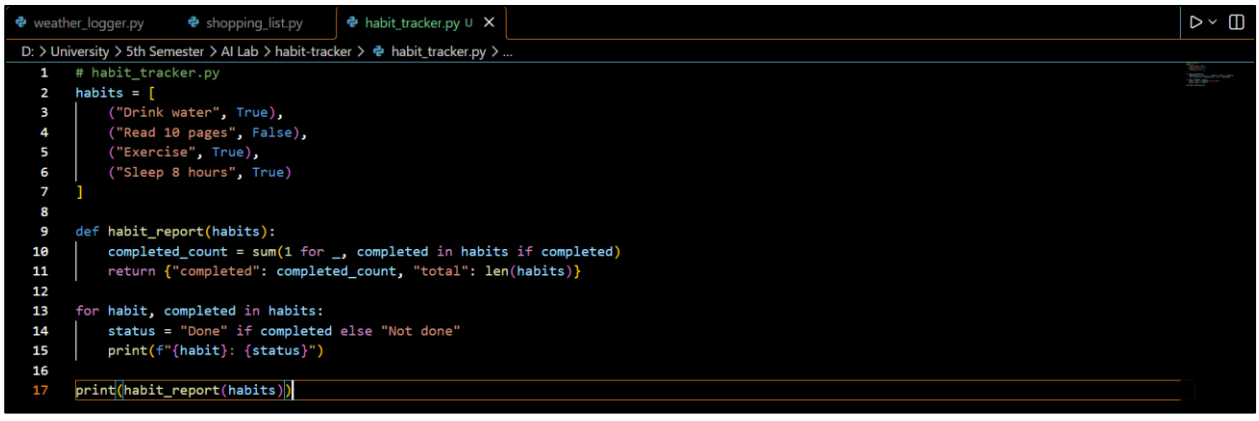
D:\University\5th Semester\AI Lab>mkdir habit-tracker

D:\University\5th Semester\AI Lab>cd habit-tracker

D:\University\5th Semester\AI Lab\habit-tracker>git init
Initialized empty Git repository in D:/University/5th Semester/AI Lab/habit-tracker/.git/

D:\University\5th Semester\AI Lab\habit-tracker>
```

Create Initial Script & Commit



Stage and commit

```
D:\University\5th Semester\AI Lab\habit-tracker>git add habit_tracker.py

D:\University\5th Semester\AI Lab\habit-tracker>git commit -m "Initial commit: habit tracker script"
[master (root-commit) 48f3d2f] Initial commit: habit tracker script
 1 file changed, 17 insertions(+)
 create mode 100644 habit_tracker.py

D:\University\5th Semester\AI Lab\habit-tracker>Step 3: Add Item, Review Diff, and Commit
```

Add Item, Review Diff, and Commit

```
D:\University\5th Semester\AI Lab\habit-tracker>git status
On branch master
nothing to commit, working tree clean

D:\University\5th Semester\AI Lab\habit-tracker>git diff
diff --git a/habit_tracker.py b/habit_tracker.py
index 32665e5..8e57d75 100644
--- a/habit_tracker.py
+++ b/habit_tracker.py
@@ -3,7 +3,8 @@ habits = [
     ("Drink water", True),
     ("Read 10 pages", False),
     ("Exercise", True),
-    ("Sleep 8 hours", True)
+    ("Sleep 8 hours", True),
+    ("Meditate", False)
 ]

 def habit_report(habits):

D:\University\5th Semester\AI Lab\habit-tracker>git add habit_tracker.py

D:\University\5th Semester\AI Lab\habit-tracker>git commit -m "Add Meditate habit to list"
[master de1d397] Add Meditate habit to list
 1 file changed, 2 insertions(+), 1 deletion(-)

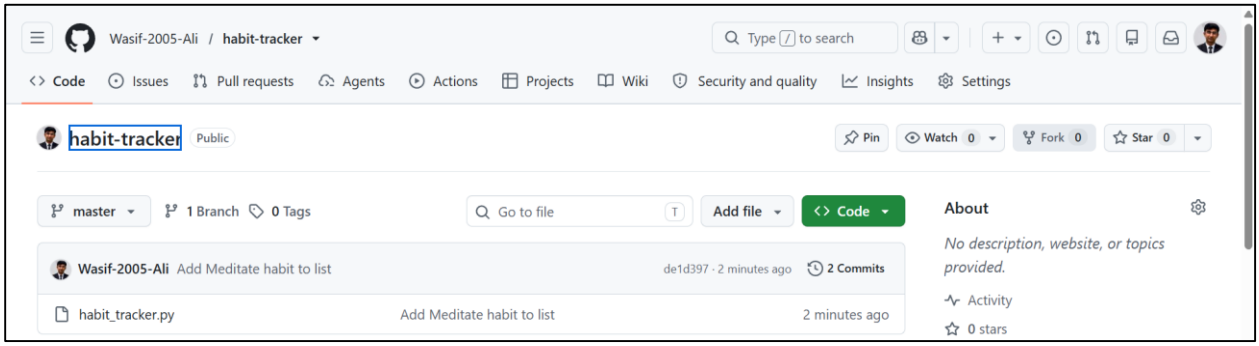
D:\University\5th Semester\AI Lab\habit-tracker>
```

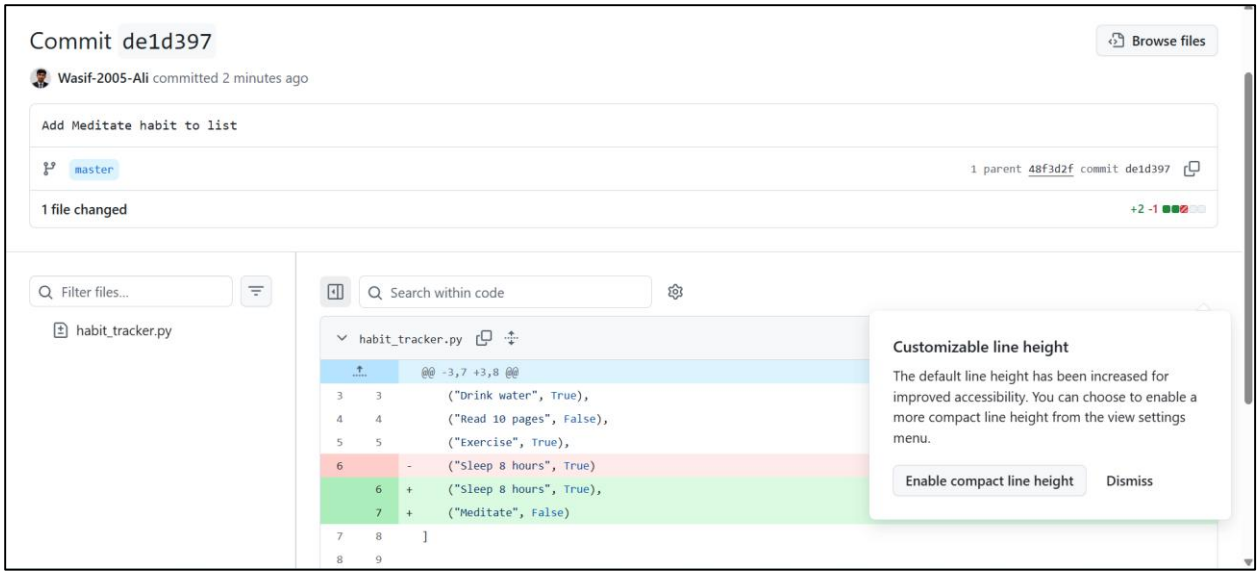
Push to GitHub

```
D:\University\5th Semester\AI Lab\habit-tracker>git remote add origin https://github.com/Wasif-2005-Ali/habit-tracker.git

D:\University\5th Semester\AI Lab\habit-tracker>git push -u origin master
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 715 bytes | 715.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/Wasif-2005-Ali/habit-tracker.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

D:\University\5th Semester\AI Lab\habit-tracker>
```





Verify Clone, Remote Edit, and Pull Workflow

```
D:\University\5th Semester\AI Lab\habit-tracker>cd D:\University\5th Semester\AI Lab

D:\University\5th Semester\AI Lab>

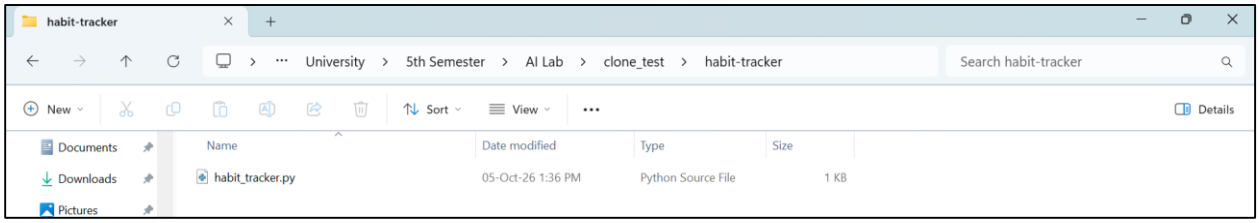
D:\University\5th Semester\AI Lab>mkdir clone_test

D:\University\5th Semester\AI Lab>cd clone_test

D:\University\5th Semester\AI Lab\clone_test>git clone https://github.com/Wasif-2005-Ali/habit-tracker.git
Cloning into 'habit-tracker'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 1), reused 6 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.

D:\University\5th Semester\AI Lab\clone_test>cd habit-tracker

D:\University\5th Semester\AI Lab\clone_test\habit-tracker>
```



!!SUCCESFULLY PULLED!!

ALL TASK COMPLETED SUCCESFULLY